Hypotheses, underdetermination, and wrong models

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Ah, but a man's grasp should exceed his reach. Or what's science for? (Anonymous)

1 How to find things out

1.1 The background: corpus-based diachronic sociolinguistic typology

This paper is about how to find things out in linguistics. More specifically, we discuss how hypotheses and theories relate to each other and what it means to *test* hypotheses using corpus data. It might seem as if the *Transactions of the Philological Society* is not be the best place to discuss the theoretical problems that we raise. However, our paper is a reply to a paper published previously in the same periodical. A journal that allows certain methods to be used in published research, we feel, should also be a place to discuss whether the methods actually work as advertised and whether they were applied accurately. The purpose of this paper is to engage the community to discuss the validity of methods embraced by an ever-growing number of researchers, and to contribute to the establishment of best-practice standards, which are urgently needed as we show by example.

The *Transactions of the Philological Society* about a year ago published Walkden et al. (2023), a paper that aims to support an established theory from the area of sociolinguistic typology. Based on Trudgill (2011) and other works by Peter Trudgill, Walkden et al. (2023, 546–47) assume that there is a fundamental distinction between long-term language contact where change is introduced primarily through bilingualism in children and short-term language contact characterised by adult L2 speakers. Details aside, there is an assumed tendency that the former leads to complexification in the affected language and the latter to simplification. Then, Walkden et al. (2023, 548–50) contrast existing approaches to support this type of theory: the *typological* approach utilises linguistic databases to examine grammatical systems in situations of language contact at the population level; the *experimental* approach uses artificial language learning situations to simulate on an individual speaker level. While both are valid and important, the *historical corpus* approach complements them according to the authors by allowing researchers to examine *when* certain linguistic changes occurred in a historical situation of language contact, and the progression of changes through the areal of the affected language should provide evidence that the change occurs predominantly and early in areas where language contact is most intense. The paper presents three corpus studies of potentially contact-driven simplification (Walkden et al. 2023, 551–61). We will argue that Study 1 and Study 2 do not—as they are—provide strong evidence or evidence at all for this hypothesis if the data are analysed with some rigour. This is due to a series of common but nonetheless insufficient ways of analysing and interpreting corpus data that we have encountered over and over in contemporary corpus linguistics. As we will hopefully make absolutely clear, our critique is directed at such methods per se. The authors actually deserve praise for making their research data freely available for reanalyses like this, a practice that is not at all common even today.

We feel a disclaimer is in order. While the general advice is often *not* to write papers like the present one because they might be perceived as destructive, overly critical, and the sign of a cantankerous character, we decided to write this paper nonetheless. In fact, the situation is ideal for a critique of methods and general practices precisely because we do not work in any of the relevant fields (sociolinguistics, historical linguistics, dialectology) and we have never had any academic or personal contact with any of the authors or their co-workers.[1] Therefore, we can credibly say that the exclusive purpose of this exercise to improve the overall quality of research in corpus linguistics of any type, including the quality of our own future work—as we certainly have learned from the re-analysis of the data.[2] We do not make any claims regarding the plausibility of the hypotheses put forward in Walkden et al. (2023), and we sincerely wish that further work provides reliable evidence supporting them. However, the paper under discussion does not.

Our discussion proceeds as follows. First, we very briefly discuss echelons of hypotheses and underdetermination in science (Section 1.2) as well as the nature of hypothesis testing and traditions of statistical inference (Section 1.3) This introduction is not intended as an authoritative or exhaustive summary of the history of the philosophy of science or modern statistics. Rather, we included it in order to position ourselves at a certain coordinate in the epistemological landscape. An important aspect of our

[1]We do work in the vast field of corpus linguistics but this is what qualifies us to write this paper in the first place.

[2]Please also note that, in order to just critique the analysis, less words would have sufficed.

claim is the use of statistical models in the paper, and we then turn briefly to statistical models in Section 1.4, connecting them to the previous discussions about hypotheses and hypothesis testing. In Section 2, we re-analyse the data sets of Study 1 and Study 2 from Walkden et al. (2023) before summarising our critique and proposing elements of a culture of sound data analysis in linguistics and philology in Section 3. Hence, readers only interested in the reanalysis of the data from Walkden et al. (2023) might skip to Section 2.

1.2 Families of hypotheses and underdetermination in science

1.2.1 Testing theories

Finding things out usually involves *hypotheses* about the world (in the broadest sense), possibly collected in coherent *theories* and either derived from observation (*induction*—if induction is possible at all) or postulated within or derived from theories and then tested against observation (*deduction*). (Or so the story goes.) Any undergraduate-level introduction to the philosophy of science will tell you that falsifiability is not a sufficient but most likely a necessary criterion for hypotheses to be taken as scientific.[3] However, we follow the broad rationalist tradition which maintains that scientific theories are models of some true reality that need to undergo continued serious testing in order to be established as adequate, a process in which we repeatedly get the chance of learning something from being wrong (Mayo 1996; Mayo 2009). A general problem in this approach is that most empirical studies attempt to test some specific hypotheses in order to corroborate a theory, but the link between the particular hypothesis (to be tested) and the overall theory is mediated by a complex network of hypotheses of different nature and granularity. Without diving too deeply into philosophical discussions, we now point out some obvious differences between the granularity of hypotheses involved in work like Walkden et al. (2023). We do so by reconstructing an approximate network of hypotheses underlying the paper, then relating that network to philosophical problems of scientific underdetermination.

Already on the first page of the paper, the authors state that the overarching purpose of the paper is to test the high-level theory proposed in Trudgill (2011), at least if *assess* is to be read as *evaluate*. Hence, we assume they're on the same page with respect to the broad philosophical approach. Highlighting was added by us.

> This article is about how to **assess Trudgill's theory** using the most powerful methodological tool that contemporary historical linguistics has at

[3] Chalmers (2013) is highly recommended as a first textbook on the philosophy of science. See Popper (1962); Lakatos (1970); Laudan (1983) for more in-depth discussions of falsifiability and the demarcation problem. The falsifiability of theories is a much more heatedly discussed topic than the falsifiability of concrete hypotheses.

> its disposal: the study of the historical record itself, in the form of textual corpora. (Walkden et al. 2023, 546)

1.2.2 Substantive hypotheses and their auxiliaries

We assume that the theory to be evaluated collects **substantive hypotheses** (the *tenets* of the theory) such as:

- Adults learn second languages through a different mechanism than children.
- The adult mechanism leads to simplification because of—ideally spelled-out—underlying cognitive principles.
- The social parameters of the contact situation influence the direction and pace of the change.
- This leads to different temporal scales at which the contact-induced change takes place.

If the axioms were refuted repeatedly by data, the theory would be abandoned. However, refuting such broad hypotheses is quite difficult. They rest on a cushion of auxiliary hypotheses, which can always be blamed for apparent refutation. Also, they are almost never tested directly, but they have to be broken down into particular hypotheses (see below), which in turn are often converted into statistical hypotheses.

The substantive hypotheses often **auxiliary hypotheses** to be true, and it is not always clear whether these hypotheses belong to the tenets or are fundamental auxiliaries. Some potential auxiliary hypotheses come to mind:

- L2 learners may exert a linguistic influence on L1 speaker of the same language.
- Language contact can induce language change under certain—ideally fully spelled out—conditions.
- The theory applies universally because it is based on general cognitive principles.
- "*Population-level patterns are not entirely random and [...] individuals' usage can be non-categorical but still pattern meaningfully in probabilistic ways*", a very specific and strong hypothesis directly from Walkden et al. (2023, 549).
- The loss of number inflection on pre-nominal elements is simplification.
- The addition of pronominal subjects in a predominantly pro-drop language is simplification.
- The move from synthetic case inflection to adpositional case marking is simplification.

1.2.3 Particular hypotheses and their auxiliaries

The authors move on to introduce the—hitherto neglected—*historical corpus-based approach*, which they claim is suitable to collect evidence for the theory. This introduces an additional tier of hypotheses, which the authors acknowledge themselves by stating that the theory makes **predictions**. The concrete predictions cannot be part of the core hypotheses, otherwise they would not be predictions but rather substantive hypotheses. The approach to these hypotheses is presented as *confirmatory* as the studies are intended to present *support*.

> In **support of our approach**, we present three case studies of morphosyntactic phenomena for which the theory of sociolinguistic typology makes predictions, and assess them using historical corpus evidence. [The three case studies] taken together [illustrate] how historical corpus evidence can be brought to bear on questions of sociolinguistic typology. (Walkden et al. 2023, 547)

Typically, predictions take the form of **particular hypotheses**. Particular hypotheses should have at least some chance of being tested using data, while substantive hypotheses are often too abstract to be tested directly. Such particular hypotheses are—in the case at hand—related to the diachronic development of:

- the loss of number marking on pre-nominal elements in English under Norse contact (Section 2.1 below);
- the rise of overtly expressed pronominal subjects in Latin American Spanish under contact with African slaves who had to learn Spanish rather quickly given the social parameters of the contact situation (Section 2.2 below);
- the reorganisation of the case and adposition systems of Slavic languages in the diffuse contact situation of the Balkan sprachbund.

These hypotheses come with their own **auxiliary hypotheses** related to their measurement and operationalisation (in this case under the historical corpus-based approach), which are not derived from any other hypotheses but rather practical requirements for the measurement procedure to be appropriate:

- In Study 1, the L1 of the conquered English was influenced by the conquerors' L2 acquisition of English through the same mechanisms as the Spanish L1 of the slave-owners was influenced by their slaves' L2 acquisition of Spanish despite the difference in social parameters. This mechanism is the same as the one at work in the Balkan sprachbund where the joined influence of Albanian, Turkish, Romance, and other languages on Slavic led to the simplification of the case marking system under rather different and complex political as well as social constellations.
- The mechanisms described by the substantive and auxiliary hypotheses are mirrored reliably and with negligible temporal delay in written sources.
- The sources available are rich and reliable enough to mirror subtle diachronic developments.

- The change in the observed cases is due to a large extent to language contact and not to overall tendencies/other mechanisms of change that happen to happen in certain areas and at a certain time.
- The authors of the text were connected enough with the relevant language community (rather than being silent monks or ivory-tower intellectuals) to be good exponents of the change under examination.

1.2.4 Networks of hypotheses, error, and underdetermination

The connections between such families of hypotheses—especially taken as causal connections—are complex and delicate. In the case at hand, a test of particular hypotheses is attempted to deliver support for substantive hypotheses, for all practical purposes assuming that hypotheses in the auxiliary families are all correct. However, the auxiliary hypotheses are apparently not understood as being implicitly tested by the procedure, and any false hypothesis in those families has the potential to mar any inferences with respect to the particular and hence the substantive hypotheses. For example, the corpus method might be completely inadequate given the available resources to the effect that we get a completely random pseudo-confirmation from the data (for example because of the writers of the available texts were not sufficiently connected within the community of language users or any other of a million reasons). Alternatively, Trudgill's theory might be incorrect. At the same time, the loss of number inflection on pre-nominal elements might not even be simplification in a relevant sense. In short, the results from the corpus studies only have the advertised interpretation if all the auxiliary hypotheses are *true to a satisfying degree*. It is clearly not enough if they are *plausible* since plausibility is only a measure of our belief in the hypothesis. For the inferences to be valid, our belief is not essential. Notice in general that a direct test of substantive hypotheses is almost certainly not within our grasp, and we have to consider what kind of evidence testing particular hypotheses is able to provide for them at all.

The concrete examples given above and the exact delineation of the families of hypotheses might deserve some refinement by researchers who are better versed in diachronic sociolinguistics, but our point is more general in that confirmation and severe testing are very hard to come by in the face of the underlying theoretical complexities of a corpus study. In epistemology, such problems have been discussed at great length, and the problem is known as the *underdetermination of science*. Pierre Duhem claimed that any disagreement with the hypotheses found in an experiment was the sign of a disagreement with a large logical conjunct of hypotheses rather than just the specific hypothesis of interest at hand (Duhem 1914). While he made an argument exclusively about physics, Willard Van Orman Quine extended this skepticism towards our knowledge or rather our beliefs in general (Quine 1951, 1955). This form of *holistic underdetermination* has led to far-reaching discussions about the validity of the scientific method. Laudan (1990) concludes that it was taken out of proportion inasmuch as we could in most cases establish reasonably well the actual cause of an experiment being in disagreement with the predictions. Arguably, however, this is not at all trivial with linguistic hypotheses and the ways we use to test them, as we are going to show in Section 2. This is because of their built-in underdetermination, which largely due to their blatant imprecision and most linguists aversion to forming theories that make numerical predictions. Linguistic are often not formalised and consequently do not make neat predictions especially for corpus data (see Section 3), we interpret (weak) accordance with the expectations as corroboration. This invites back all problems of holistic underdetermination with a vengeance, as any of the families of imprecise and often implicit hypotheses (see above) might conspire to result in a very loose fit of the data to the theory, which in the worst case is taken as confirmation. Metaphorically speaking, a circle can be drawn around any geometric figure, as long as it is large enough.

Another often discussed for of underdetermination is *contrastive underdetermination*. Contrastive underdetermination describes the fact that there might be or actually are potentially many alternative theories that could explain the same data. Laudan and Leplin (1991) dismiss contrastive underdetermination based on—among other things—the fact that it is unlikely that two theories receive exactly the same amount of empirical support from some data despite both being compatible with the data. For linguistics, we consider this kind of underdetermination to be swamped by holistic underdetermination. Many of our theories are so imprecise and pre-formal that data (especially quantitatively measured data) cannot decide between two theories.[4] For example, the loss of plural marking on all elements but the head noun in an NP could also just accidentally coincide with the Viking contact situation. It could be a consequence of the obvious overall tendency of English to lose most of its inflectional markers from Old English to Early Modern English. If the change progresses from the North to the South during the time of the Danelaw, the sociolinguistic theory would receive somewhat better support than the theory of accidental coincidence, but the data would still be compatible with both theories.

As a consequence, linguistic theories are rarely refuted by data. While many linguistic theories went out of fashion, we ask readers to name one major linguistic theory from between 1990 and now that was made untenable because it was in clear and repeatedly demonstrated discord with data. Compare this to, for example, Newton's theory of gravity, which was ultimately refuted by anomalies in the precession of Mercury's

[4]Some rare examples of precisely formalised theories are model-theoretic semantics in the tradition of Montague (1973), model-theoretic syntax in the tradition of Pollard and Sag (1994), or proof-theoretic syntax with model-theoretic semantics as in Carpenter (1997). However, such truly formalised theories are not easy to support or refute empirically either, among other things because they, too, do not make numerical predictions about human behaviour.

perihelion going back to observations by Le Verrier (1859). The numbers from the theory and the observations simply did not match up, and there was no way around that. We also challenge the reader to imagine any obtainable set of data that would conclusively rule out all other (reasonable) explanations for the hypotheses put forward in Walkden et al. (2023). All we seem to be able to expect from corpus studies like the ones we customarily conduct is: (i) either the study does not contradict the logical conjunct of our hypotheses (which is *not* confirmation); (ii) or it is not compatible with that conjunct or our interpretation of any of the conjoined hypotheses (which is *not* refutation). Because of their loose-fit character and the holistic and contrastive underdetermination in linguistics, we can seek neither confirmation nor refutation for substantive linguistic hypotheses. This is an important point to keep in mind when we examine the role of statistics in testing hypotheses in the next section.

1.3 Statistical hypotheses and severe testing

For quantitative data-based studies, it is necessary to derive **statistical hypotheses** from particular hypotheses described in the previous section. A statistical hypothesis formulates the expectation of a numerical outcome in an operation of counting or measuring in a corpus study or an experiment, etc. Du to the aforementioned low degree of formalisation, a *numerical outcome* is usually something as imprecise as *n should be greater than m* or *n should be less than or equal to m*. For example, with regard to the loss of number marking on pre-nominal elements in English under Norse contact, we could derive the expectation that the number of occurrences of determiners and adjectives without overt number marking should be higher in Northern and Eastern English dialects compared to Western and Southern dialects during the period of the Danelaw because of the assumed underlying mechanisms of sociolinguistically induced language change. Let's abbreviate this hypothesis, drop some understood conditions, and call it H_1: *Occurrence counts of inflected pre-nominal elements are* ***lower*** *in the North and East compared to the South and West.*

Once we have got statistical hypotheses, it is finally the moment of truth. By measuring and counting, we obtain numbers from our samples, and we check how well the numbers are in accord with certain statistical hypotheses. Because of some statistical intricacies, which cannot be detailed here for space reasons, it is difficult to check for *accordance* of data with hypotheses, and it is much easier to check for divergence from hypotheses. Hence, one inverts the original hypothesis, generating a so-called *Null hypothesis* or just *Null.* H_0: *Occurrence counts of inflected pre-nominal elements are* ***equal to or higher than*** *in the North and East compared to the South and West.* The theory of frequentist statistics then allows us to calculate the (pre-experiment) probability of obtaining a sample that diverges as extremely as or more extremely than the one actually obtained if the Null were true: the p-value. If the event that actually occurred had (past tense!) a very low probability under the Null before the experiment was conducted (and the p-value is thus low), one usually proceeds to a positive inference regarding H_1. Correct interpretations of such results are a delicate matter, especially for practitioners with suboptimal training in statistics.[5]

Mostly unnoticed in the linguistics community, such inferences have been under heavy debate in the past twenty years.[6] Routinely, inferences are made using Null-Hypothesis Significance Testing (NHST), a failed system of deciding which measured statistical results are *significant*, and which are not (for how and why it failed see Gigerenzer 2004; Ioannidis 2005; Colquhoun 2014; Perezgonzalez 2015; Greenland et al. 2016; Munafò et al. 2017; Mayo 2018). NHST works like this: Significance is reached when the p-value is lower than some pre-set threshold (often 0.05), and this is taken as straightforward corroboration of H_1, which is more often than not incorrect. NHST is usually understood as a distorted hybrid of the philosophies of inference by Ronald Aylmer Fisher (Fisher 1935, 1959) on the one hand and Jerzy Neyman and Egon Sharpe Pearson (Neyman and Pearson 1933; Neyman 1937, 1957) on the other hand.[7] Under the *Severity* approach from Mayo (1996); Mayo and Spanos (2006); Mayo (2018), NHST does not meet the requirements of a *severe test* of a hypothesis. The *Severity Principle* is defined as (in its weak form and adapted by for readability): *Some data do not provide good evidence for a hypothesis if they result from a test procedure with a very low probability or capacity of having uncovered the falsity of the hypothesis, even if it is incorrect.* (paraphrased from Mayo and Spanos 2006, 162). We would like to posit the banality that a theory can only be called *confirmed* or even just *corroborated* must have survived at least a few severe tests. If a test is interpreted as straightforward corroboration of a complex network of hypotheses (see Section 1.2) simply because some minor divergences from a single derived Null hypothesis were detected in a (possibly quite large) sample, it should be considered a blatant violation of the Severity Principle. Such tests have an amazingly weak capability of detecting any error in substantive or even in particular hypotheses.

The most important point of critique for our purposes is indeed that even minor divergences (even if they're actually there) make the Null false, which is easily detected in NHST in the form of a low p-value. This is not a flaw in the foundations of frequentist inference but in its application. Think about it: Even if the effects postulated in Walkden et al. (2023) only accounted for the behaviour of 0.001% (one ten-thousandth) of all speakers, the Null hypothesis would technically be false. Still, the Null being false

[5]Our overall introduction to frequentist statistics can be verified using any sound text book on statistics. We recommend Vasishth and Broe (2011) for beginners, and we're also in the process of writing our own open-access textbook on the matter.

[6]There is, however, a debate about NHST in corpus linguistics independent of the mainstream discussion, for example in Kilgarriff (2005); Koplenig (2019).

[7]An erudite comparison of the two including biographical details is Lehmann (2011).

by such a small numerical margin would not matter much for our understanding of the changes in the English language under the Danelaw. Despite its theoretical relevance, NHST is perfectly able to detect this minor discrepancy if the sample is large enough, and we could take this erroneously as a corroboration of Trudgill's theory. However, we clearly should *not* take it as corroboration as we did not perform a severe test of the theory.

We see many of the deficiencies in Walkden et al. (2023) to be detailed below as stemming from an adherence to NHST. While proposed solutions to the NHST crisis have been everything from radical to technically demanding, we will suggest in Section 3 that some very straightforward principles might have avoided these fallacies of NHST in the paper: (i) Researchers must examine their data and their models carefully without confirmation bias. They should not make bad NHST inferences, and they should receive proper training in the fundamentals of applied statistics. (ii) Editors and reviewers must receive as much relevant information as possible in order to spot problems with statistical models. Reviewers and editors must be competent to spot errors in the application of statistical and data-driven methods. (iii) The burden of reporting only positive findings in the most optimistic tone must be lifted from researchers, above all through pre-registration. Otherwise, the whole field of corpus-oriented linguistics and philology is prone to the detrimental effects of data mining and data dredging. While we do not accuse the authors under discussion of either data dredging or even negligence, the whole field will suffer if publications like Walkden et al. (2023) get published without any substantial methodological critique. On the contrary, we are perfectly certain that the authors acted in good faith, as we have not only observed similar confusions in other researchers over and over again, but also in our former selves.

1.4 Statistical models in corpus linguistics

As the final prerequisite for our discussion of the errors in Walkden et al. (2023), we now turn to the specific statistical tools used in the paper. The authors utilise statistical models of a certain kind, and a few words on these models are in order, above all connecting them to the previous discussion about hypotheses and hypothesis testing. Maybe no other advanced statistical method has been popularised more in the past twenty years than *Generalised Linear Models* (GLMs). A highly influential paper in the popularisation of GLMs was Bresnan et al. (2007), quickly followed by Baayen (2008), which provided an accessible introduction to GLMs for linguists in a textbook, Gries (2013) being another textbook that included GLMs. A few years later, Gries (2015) still called Generalised Linear *Mixed* Models (GLMMs)—an extension of GLMs to be discussed presently—underused, but it is fair to say that diverse variants of GLMs now (ten years later) belong to the repertoire of statistics for corpus linguists. GLMs and their variants are related to regression, and they have a very simple purpose: they estimate how the variance in a set of variables (the *regressors*, *predictors*, *explanatory variables*, or *independent variables*) is linked to the variance in another variable (the *response*, *outcome*, or *dependent variable*) and *explains* the variance in this response variable. Regressors might, for example, be the animacy of a dative object in an English sentence (a binary variable with potential values *animate* and *inanimate*) or the length of the object NP. A possible response variable could be whether the dative shift construction was used in the sentence or not.[8] In other words, the regressors *vary* (not all instances have the same value for *animacy* and *length*), and depending on their concrete values in a given sentence, the choice of the the response variable (dative shift or no dative shift) can (hopefully) be explained and thus predicted.

One would not expect to be able to predict all cases of dative shift correctly, but by estimating the influence of the two regressors one would certainly hope to make better predictions than by guessing. Statistical models allow us researchers to specify which regressors we expect to exert an influence on any variation phenomenon. Then, the *estimator*—a piece of maths usually implemented in software—uses existing data such as thousands of sentences with dative objects for which we have annotated the observed values of the regressors to provide the best possible numerical estimates of their influence on the response.[9] Often, the mere quantification of the influence exerted by the various regressors on the response is more valuable to researchers than the ability to make actual predictions of what some speaker of English might say or write in the next utterance.[10] In order to understand how language works, we desire to know what influences the choices that speakers make, and how strongly they do so, expressed in numbers. Estimators for GLMs, GLMMs, etc. provide such numbers.

While LMs (*Linear Models*) are used if the response variable is measured on a numerical scale, and it depends linearly on the regressors, a Generalised Linear Model (GLM) uses more indirect maths to model many kinds of relationships between regressors and response variables. In linguistics, it often quantifies the influence of the regressors on a binary choice, in which case the model predicts the *probability* of one variant being chosen over the other. GLMMs (*Generalised Linear Mixed Models*) include so-called random effects. These capture variation in the response variable at the group level, for example if there are multiple measurements from the same speaker, from the same text, containing the same verbs, etc. (see Schäfer 2020 for an overview). The

[8]Indeed, the dative shift in English has been excessively researched in variationist linguistics, often using modelling techniques or regression, see for example Bresnan and Ford (2010); Theijssen et al. (2013); Wolk et al. (2013); Röthlisberger, Grafmiller, and Szmrecsanyi (2017).

[9]A very thorough introduction to the applied art of modelling is Gelman and Hill (2006). The maths of the estimators is explained very well and accessibly in Fahrmeir et al. (2013), although the adverb "accessibly" has to be evaluated relative to the complexity of the subject.

[10]Also, there are much better ways of making such predictions based on information theory today, such as large language models.

estimator uses different maths for random effects, but it is often possible to include an effect either as a normal regressor (*fixed effect*) or a random effect (Gelman and Hill 2006, 245–47).

There is a huge number of possible evaluations of the quality of the model fit. In Walkden et al. (2023), the dominant one are NHST tests of the coefficients. The coefficients are the numbers quantifying the strengths of influence of the regressors on the response. In these tests, the Null hypothesis is always that the coefficient is 0 (no influence), and the per-coefficient p-values are only correct if the rest of the model is at least approximately correct. This is a big assumption, and one should not rely on a significance threshold of 0.05 under such uncertain conditions, at least not if a severe test of any substantive hypothesis is desired. Among the many further evaluation metrics are R^2 measures. These are attempts to quantify (in some cases merely approximate) the proportion of the variance in the response that is explained by the totality of the model. In mixed models, the *conditional* R^2 includes the random effects, and the *marginal* R^2 does not include the random effects. The *prediction accuracy* is strongly correlated with R^2, and it is the proportion of correct predictions that the fitted model makes, either on the original data or on new data. The prediction accuracy is related to the *base rate*, which is the accuracy achievable by always guessing the most frequent outcome. If a language has subject drop in 90% of all cases, a very uninformative model might always predict subject drop and still reach 90% (a proportion of 0.9) base rate accuracy. However, a model that reaches 90% accuracy by differentiating between the two outcomes (instead of always predicting the most frequent outcome) is still better than a completely uninformative model at the same prediction accuracy. We'll see completely uninformative models below, however. The *Akaike Information Criterion* (AIC) is a measure based on information theory. It has no meaningful absolute numerical interpretation, but it can be used to compare models. A model with a lower AIC tends to *be better* than one with a higher AIC. There are many more evaluations and quality checks, but these basic ones suffice to show that the models reported in Walkden et al. (2023) are bad models.

We would like to highlight one further point which is to some extent merely terminological. The model itself is (or ought to be) a causal model of what controls the variance in the response. A model is thus *specified* by researchers based on theoretical assumptions or previous knowledge acquired through measurement. The estimator *estimates the parameters of the model*, above all the coefficients. Hence, we do not "run a model" in a piece of software (Walkden et al. 2023, 558). Several questions immediately emerge: (i) How do the specification of the model and the estimation of its parameters ensure that the regressors encode *causal* influences. The answer is chillingly simple inasmuch as they absolutely do not in any way ensure this. One could also (with partial success and a certain precision) model the length of an NP (measured in the number of words) based on whether it has undergone dative shift or not, completely reversing causality. (ii) How do we decide whether the estimates of the regressors' influences are actually relevant and not minor influences concerning only a small fraction of sentences in the corpus? The data fed into the estimator might be too little to reach the required representativity, or they might be biased. In corpus linguistics, biasing usually occurs because of opportunistic sampling or because too few data are supposed to cover too large a population across time and/or space. Also, there might be sampling-related quirks in the data set (such as data from a few or just one text with highly idiosyncratic properties compared to all other texts in the sample), or some effect might be really small and just relevant for a tiny subset of the data, etc. In the light of these questions, what does it mean if an estimator comes up with a number like β=0.03 (see below) for the coefficient of a factor allegedly influencing a variation phenomenon? We'll present a thorough evaluation of the models and estimates reported in Walkden et al. (2023) to show that it might mean next to nothing.

2 Data and inferences in Walkden et al. (2023)

2.1 Study 1: Number agreement in English

2.1.1 What was reported

Study 1 deals with the loss of plural agreement marking in the NP during the Middle English period. It is interpreted as a reduction of redundancy when plural markings are lost on attributive adjectives and determiners. Per hypothesis, this is partially due to the influence of Scandianavian languages from the eighth until the eleventh century. Also, the different pace by which determiners and adjectives lose their agreement markings is analysed as possibly related to language contact. The study looks at determiners (referred to as *quantifiers* throughout, although not all determiners are quantificational) and monosyllabic adjectives in 47 texts totalling 4,366 data points (called *tokens* on p. 551).

Based on a plot overlaying texts as dots on a map of England (their Figure 1), it is argued that some texts show stronger loss of inflection on the determiner than on the adjective.[11] Walkden et al. (2023, 552) claim that "this difference does not depend on the date of composition of the manuscript, but rather on the dialectal region. The majority of texts which show a difference between quantifiers and adjectives come from areas where Scandinavian contact was intense." The parameters of a linear model "with the difference in agreement between adjectives and quantifiers as our dependent variable and region as independent variable" are estimated. With the Southern

[11] We would like to point out that it would have helped to also include the borders of the Danelaw in the plot. Also, the visual pattern does not appear to be very obvious to us.

dialects as the reference category (because they had the least amount contact with Scandinavian languages), the authors find a *significant effect* for North and the East Midlands but not the West Midlands (Walkden et al. 2023, 552), where Footnote 8 was omitted:

> The results showed a significant effect of region for the North (β=0.39, SE=0.15, t=2.6, p=0.01) and the East Midlands (β=0.2, SE=0.10, t=2.09, p=0.04), but not the West (β=0.13, SE=0.13, t=1.0, p=0.3). This geographical distribution suggests that contact with Scandinavian speakers played a relevant role in the loss of inflectional morphology, but equally importantly, it shows that concord in quantifiers was more affected than concord with adjectives.

Apparently, the inferential logic is pure NHST. Any result with p<0.05 is called *significant*. We will return to these results in the next section and show why *significant* does not mean *meaningful* or *substantive*. However, there is a more fundamental problem to be discussed first. The paper is very unclear with respect to the measurement of the difference in agreement between adjectives and quantifiers as used as the dependent variable. It cannot be a direct observation because it is described as a *difference* of some kind, and it appears to be a score calculated for entire texts, probably as some kind of proportion. Luckily, the data package reveals what was actually computed. Per text, the authors calculated (i) the proportion of determiners with plural agreement, (ii) the proportion of adjectives with plural agreement, and (iii) subtracted the former from the latter. The linear model uses this difference score as the dependent variable (or response variable).

Instead of discussing problems with such an approach from a theoretical point of view, we plot the resulting data to expose them by example. Figure 1 shows all 44 texts (x-axis), sorted by their difference score from lowest to highest, with the difference scores on the y-axis. However, we added more information that is not revealed in the paper itself. The shape of the dots corresponds to the dialectal regions (see legend), and their size corresponds linearly to the total number of observations (summed counts of determiners and adjectives), a number that ranges from 12 for *Edmund-m4* to 315 for *Purvey*. The vertical bar for each dot encodes one further remarkable numerical property of the texts and their scores: it shows how differently determiners are overrepresented in the texts. It is unclear whether the counts provided by the authors are the total numbers of adjectives and determiners in the texts or whether the number represents how many were annotated. In any case, some texts contribute far less adjectives relative to the number of determiners than others:[12] Note that the scale of the bars is not the same as the scale of the y-axis and that long lines have been

[12] It is not surprising that a text contains more determiners than adjectives. The question is rather why the ratio differs so wildly between the chosen texts.

shortened to fit into the plot. Consequently, the numbers should be inspected for a full picture. *Edmund* contributes 2 adjectives and 44 determiners, which means it has 22 times more determiners than adjectives. On the other hand, *Reynard* contributes 13 adjectives and 16 determiners, which means it has 1.23 times more adjectives than determiners.

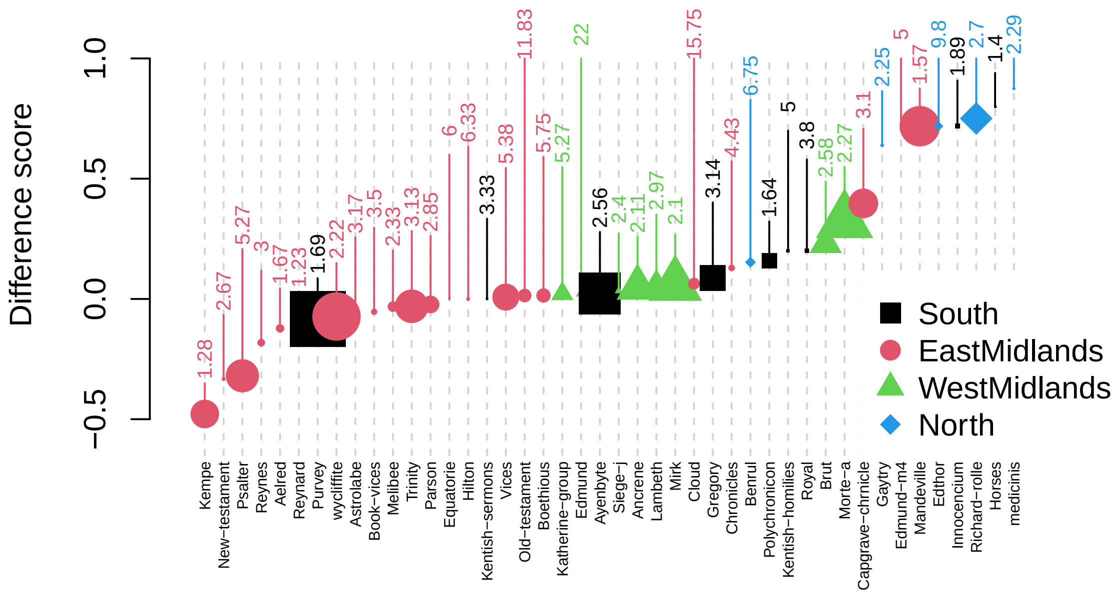

Figure 1: Difference score for the texts in the sample; sizes of the dots correspond to the number of observations contributed by the text; vertical bars indicate the overrepresentation of determiners

Using such aggregated data to estimate the parameters of a linear model is detrimental. The unweighted model treats each text as contributing the same amount of information, which simply and obviously is not the case. The established purpose of such models in empirical sciences is to model the distribution of data points, not their aggregations (see Gelman and Hill 2006 if in doubt).[13] Thus, the model from the paper makes the assumption that a text with 12 data points (the observed mininum) and a text with 315 data points (the observed maximum) can be treated as contributing the same amount of evidence. This is ill-advised for all types of models, even much simpler ones, as a fictitious example shows. Assume we had the task of calculating the average length of a newly discovered Italic language, Faloscan. There are two Faloscan inscriptions found, one containing two short sentences averaging 3.5 words per sentence because it contains two sentences, one 3 words long, the other one 4 words long.

[13]There are models for proportion data and other aggregated data. However, they are much more difficult to handle and—above all—interpret (Zuur et al. 2009). Most crucially, however, the authors report a plain linear model and *not* a weighted binomial model on proportion data.

The other one is a longer legal text comprising 25 sentences with an average length of 8.3 words per sentence. A phylogeneticist includes Faloscan in their database of average sentence lengths, calculating an average length of $(3.5 + 8.3) \div 2 = 5.9$ words per sentence. Almost anyone would criticise this as being incorrect, as the shorter text should receive a much lower weight in the calculation of the average.[14]

However, Walkden et al. (2023) do not calculate a simple average, they estimate parameters of a much more complex model based on even more extremely unbalanced aggregated metrics. For example, if *Innocendium* and *Horses* were texts as long as (or sample from texts as large as) *Purvey* and *Ayenbyte*, all generalisations about the Southern dialect group would disappear.[15] This is so elementary, that at least one of the authors, the reviewers, or the editors should have noticed it.

When we specify a model, we assume all kinds of sources of influence on variance in the regressor **per data point**, which are the actual measurements. The model is specified to relate the systematic (explained) influences (variance in the explanatory variables) to the variance in the response variable, leaving the unexplained variance as residuals. However, in the case at hand the difference score is itself an estimate based on wildly varying sample sizes and wildly varying balance between the numbers of adjectives and determiners, which means that it comes with a kind of variance—or, plainly, inaccuracy—that the simple linear model and its interpretation are not made for (in the mathematical sense). This also means that actual significance levels very likely do not correspond to nominal significance levels, and at least using a more conservative significance level than 0.05 would be recommended. In the next section we report an alternative model of the data from Walkden et al. (2023), and we show what remains of the results advertised in the paper under more accurate modelling.

2.1.2 Alternative models

Luckily, the original unaggregated data are available in the data package. Now, even in the middle ages, the relevant linguistic event was the actual production of a single instance of a determiner or an adjective by a writer. Therefore, the best model in the case at hand is one that estimates the probability of speakers overtly marking plural agreement in a concrete sentence or not.[16] We specified a model that comes close to the idea in the paper but which avoids the problems with the unbalanced difference

[14]The correct calculation of the weighted mean in this case is $(0.075 \cdot 3.5) + (0.925 \cdot 8.3) = 7.94$ words per sentence.

[15]We would like to stress that it is irrelevant whether more data has been annotated in the meantime and whether the picture has become clearer, changed, etc. Walkden et al. (2023) is published research that has undergone peer review, and it is this specific paper we are looking at from a methodological point of view. It has been downloaded 1,440 times as of 29 March 2025.

[16]Inconsistency, the authors follow such a strategy in Study 2, see our Section 2.2. However, they use another incorrectly aggregated score as a regressor in that study.

Table 1: The Paper model is a binomial model with a logit link. N=4366, AIC=4681.79, Tjur's R^2=0.1473. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.75 given a base rate of 0.71.

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Adjective, South, 0)	1.96	10.72	<0.0001	***
POS	Determiner	-0.38	1.78	0.0752	
Dialect	WestMidlands	-0.72	3.40	0.0007	***
	EastMidlands	-1.53	7.02	<0.0001	***
	North	-0.53	1.55	0.1216	
POS:Dialect	Determiner:WestMidlands	0.52	2.12	0.0344	*
	Determiner:EastMidlands	0.08	0.33	0.7388	
	Determiner:North	-2.52	6.47	<0.0001	***

scores is the following one. It models the probability of the realisation of agreement on a word as a function of is part-of-speech (adjective or determiner), the dialect group, and an interaction between the two. We call it the *Paper model* even if it is not what *was* reported in the paper but what *should have been* reported in the paper. See Table 1 for a report of the coefficients and the associated p-values as found by the estimator.

The parameter estimates indicate that (assuming a cavalier NHST significance level of 0.05 for the sake of the argument) that writers from the Midlands (East and West) have a stronger tendency to drop agreement in general, but that writer from the North do not (see the negative coefficients and the low p-values). However, writers from the Northern dialect seem to have a relatively strong tendency to avoid agreement specifically on determiners (see the interaction between POSDeterminer and DialectNorth). The last point is in line with the claims in the paper. However, the West midlands even have the stronger tendency than the East Midlands to avoid agreement overall (including on determiners). This runs contrary to the claim in Walkden et al. (2023).

In all fairness, if we drop the influence of dialect from the model specification, we arrive at a model that—according to standard metrics like R^2—is actually worse, which might indicate that the regressors do indeed have some influence on the response. We call it the *Null Model*, see Table 2.

With all this, we should keep in mind that the plot in the previous section shows that there simply are not that many data from the North. The fact that DialectNorth by itself does not reach the nominal significance level in the GLMs could also be due to the fact that there simply are not enough data points from the North and that the effect itself is, at the same time, not verty large. Also, the plot shows that individual texts and their dialect group are all over the place in terms of ther difference score,

Table 2: The Null model is a binomial model with a logit link. N=4366, AIC=5260.12, Tjur's R^2=0.0047. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.71 given a base rate of 0.71.

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Adjective)	1.15	16.86	<0.0001	***
POS	Determiner	-0.35	4.53	<0.0001	***

Table 3: Model comparison for the Null model

Name	AIC	R2_Tjur	RMSE
AgreementNullModel	5260.122	0.0047341	0.4538269
AgreementPaperModel	4681.792	0.1473388	0.4200577

and most texts have a difference score very close to 0 anyway. At the negative end, for example, there are merely five texts from the East Midlands that account for all the negative values. Hence, it seems appropriate to model the variance at the document level as a random effect, accounting for the different circumstances under which the documents were written, register differences, differences in individual styles, etc. The resulting model is reported in Table 4.

Immediately, we see that the variation between the texts swamps all other explanatory factors while the overall explanatory power of the model goes up considerably, as measured in the conditional R^2=0.578 compared to Tjur's R^2=0.147 from the Paper model.[17] The only relevant generalisation that survives at the nominal significance level is the interaction between POSDeterminer and DialectNorth, i.e., writers from the North avoid agreement on determiners more strongly than on adjectives. It should be kept in mind that this generalisation is based on 350 data points from 5 texts spanning a period of 340 years. Regardless of nominal significance, a careful interpretation seems wise. While it is sometimes necessary to work with such tiny amounts of data in historical philology, it is questionable whether statistics are the right tool. We proceed to a further inspection of the models.

2.1.3 Interpretation of the models

A (Generalised) Linear (Mixed) Model can be used to make predictions and thus go beyond any testing framework, be it NHST or any more correct inferential framework. In the case of Binomial models, it can estimate the probability with which the event

[17]While R^2 measures have been criticised and cannot always be compared between different types of models and types of R^2, this difference is so considerable that denying it is futile.

Table 4: The Text model is a multilevel binomial model with a logit link. N=4366, AIC=3202.59, conditional R^2=0.5781, marginal R^2=0.1546. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.86 given a base rate of 0.71. The only random effect is Text with 46 levels (s=3.3).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Adjective, South, 0)	1.38	2.08	0.0374	*
POS	Determiner	-0.50	1.98	0.0473	*
Dialect	WestMidlands	1.13	1.45	0.1458	
	EastMidlands	0.29	0.31	0.7576	
	North	1.09	0.97	0.3301	
POS:Dialect	Determiner:WestMidlands	0.16	0.55	0.5800	
	Determiner:EastMidlands	-0.63	1.95	0.0517	
	Determiner:North	-3.80	7.36	<0.0001	***

(in this case overt realisation of a subject pronoun) occurs. If it estimates a probability greater than 0.5 for some combination of regressors, it predicts the occurrence of the event, otherwise it predicts its non-occurrence. This probability threshold of 0.5 is successful if the model is approximately correct, but under some extreme conditions, it can be better to adapt it to another value between 0 and 1. We now check in tabular form what binary predictions the model makes on the 3,773 observations it was estimated on with a threshold of 0.5. Note that it should perform particularly well on those data. New data would always be harder to predict correctly for any statistical or machine-learning model.

Table 5 summarises the predictive accuracy of the *Paper model* (which is the GLM that implements the idea from the paper in a mathematically more sound way). The columns correspond to the actual cases of overt an non-overt agreement in the data, the rows tabulate the decisions based on the model estimates against those. The predictions are correct in 3259 cases and incorrect in 1107 cases, i.e., 74.6%.[18] This is slightly better than the base rate (proportion of the modal category, i.e., the percentage of cases in which agreement is overtly realised) of 70.8%.[19]

The *Null model* (not shown in tabular form) with only part-of-speech as a predictor is considerably weaker inasmuch as it always predicts overt agreement and hence does not differentiate at all. However, the *Text model* (Table 6), which is the one that

[18] R^2 coefficients correspond with the prediction accuracy, and the two convey a similar notion of the adequacy of the model given the data.

[19] The base rate is relevant inasmuch as one can usually make above 50% correct predictions simply by always guessing the modal—most frequent—outcome. In the long run, the correct guesses then equal the base rate.

Table 5: Correct and incorrect predictions of the Paper model

	Agreement	No Agreement
Agreement	3038	1056
No Agreement	51	221

Table 6: Correct and incorrect predictions of the Text model

	Agreement	No Agreement
Agreement	2856	393
No Agreement	233	884

incorporates a random effect for text, does much better than anything reported in the paper. Its predictions are correct in 3740 cases and incorrect in 626 cases, i.e., 85.7%.

As mentioned above, sometimes models make biased probability estimates, and an adapted threshold (for example, lower than 0.5) for binary decisions leads to better results. This is often the case in rare events regression (King and Zeng 2001), which is used when the base base rate is even higher (for example, around 99%). Adjusting the threshold at which we make the binary decision incrementally from 0 to 1, we arrive at Figure 2 for the Paper model (left panel) and the Text model (right panel). While it is not a model that differentiates very well, we see that the Text model is at least able to differentiate beyond the mere prediction of the modal category (and thuis landing at the base rate). The peak is close to 0.5, as expected for normal binomial models. The Paper model essentially predicts at the level of the base rate all the way until a threshold of roughly 0.8 is reached, at which point the prediction accuracy drops dramatically to 100 minus the base rate.

Finally, we can approach the accuracy of the model from two other perspectives. The *Precision* is the proportion of correct predictions of the event occurring (here, the subject being overtly realised) among all predictions of this event. In other words, it is the number of true positives divided by all positives (true or false). The *Recall*, on the other hand, is the proportion of correct predictions of the event occurring among all the cases in which the event actually occurred. Put differently, it is the number of true positives divided by the number of events that have occurred. These two scores, evaluated at different thresholds, can be used to assess the quality of the model and adjust the threshold. The balanced point (where Precision and Recall reach their highest values and the curves cross) is around 80% for the Paper model and around 90% for the Text model, which is another indicator that the Text model differentiates much better.

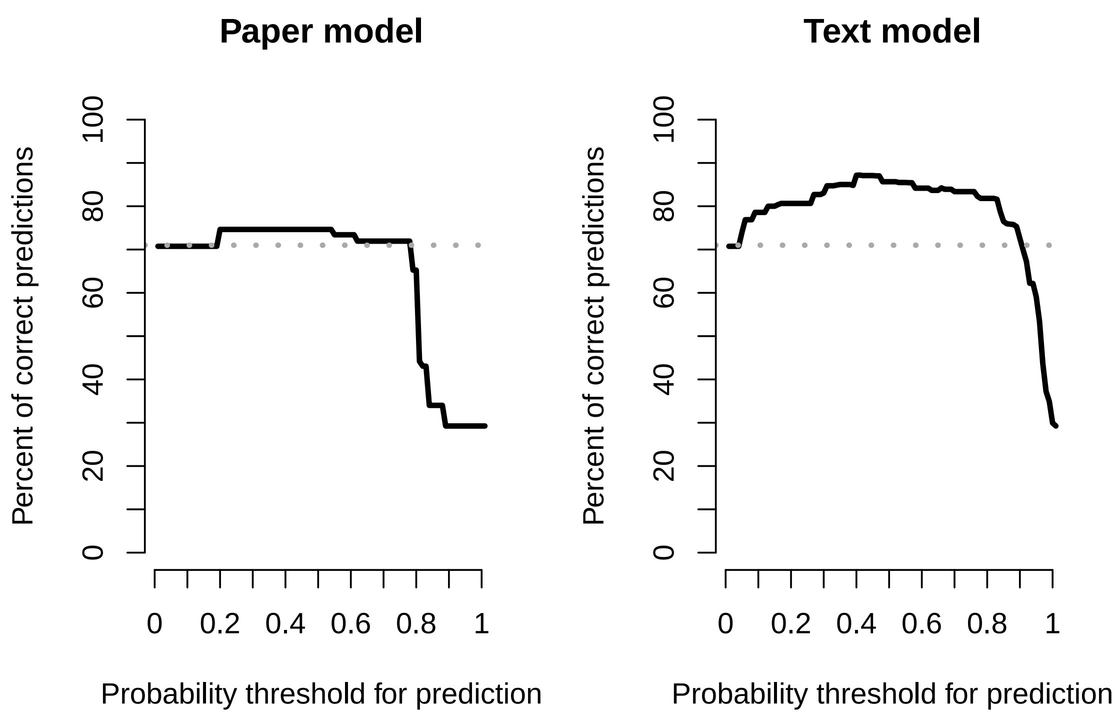

Figure 2: Percent of binary predictions based on the Paper model and the Text model at varying theresholds; only for data used to estimate the model parameters and grouped by their actual outcome

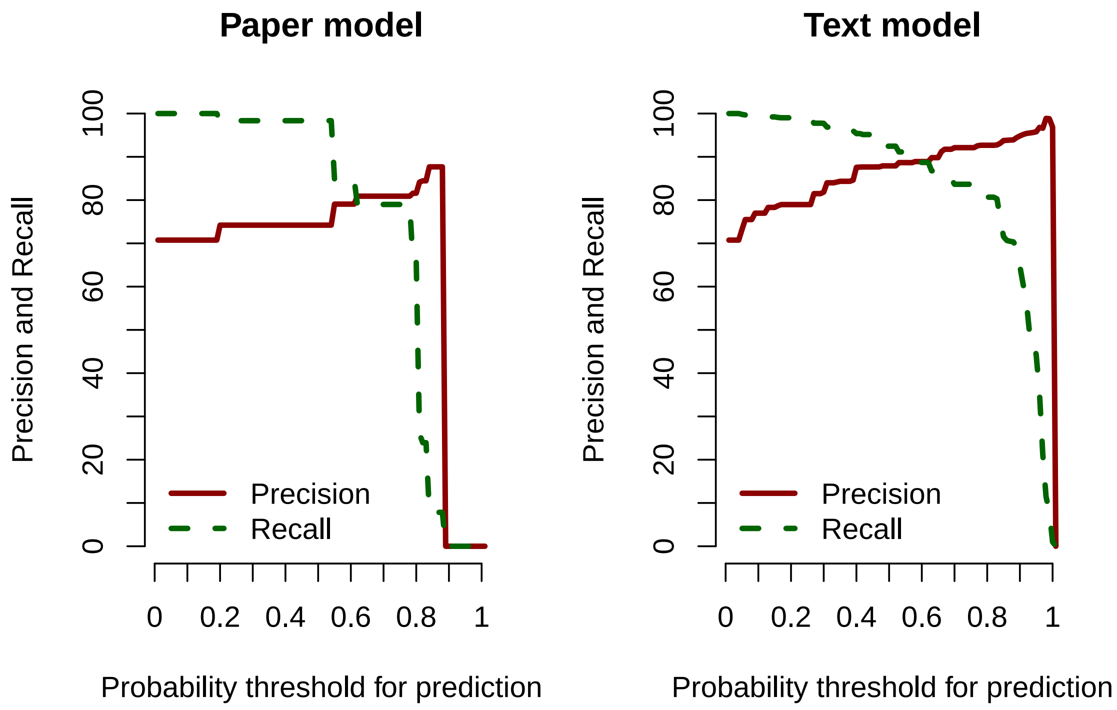

Figure 3: Precision and Recall for the Paper model and the Text model at varying theresholds; only for data used to estimate the model parameters and grouped by their actual outcome

What does all this mean? It means that the inferences in the paper are wrong, and that the data in Study 1 do not provide evidence in favour of the network of hypotheses explicitly or implicitly introduced (see Section 1.2), even if the model is reformulated without the inadmissible aggregation in the form of the difference scores. The influences of the regressors on the choice between overtly marked and unmarked prenominal elements are so minor that only typical NHST misinterpretation of p-values could wrongly detect them as having a relevant influence. What is more, the dialectal region has minor and above all contradictory pseudo-influence inasmuch as there is no clear North-East to South-West continuum. If the individual text is added as a random effect, we see that the single influence with at least moderate explanatory power is the individual text (and thus likely a combination of style and register), not the dialectal region. It should be noticed that including the text as a random effect is not even an optional decision as *not* including it violates assumptions underlying the statistical model, in this case predominantly the *independence of events*. To summarise, the data from Study 1 corroborate an old wisdom from traditional philology, videlicet that texts differ and deserve careful individual examination in order to be decoded and understood properly.

2.2 Study 2: Null subjects in Spanish

2.2.1 What was reported

In Study 2, the authors set out to model the influence of orality, time, and region (Spain versus Latin American countries) on the tendency to use overt subjects in Spanish despite the overall very strong tendency towards subject drop in Spanish. Orality is measured as the ORScore, which appears to be an aggregate relative frequency of certain grammatical features in the respective texts. It is thus a derived construct and not a direct observation. Given the fact that it is composed of proportions (relative frequencies) the warnings raised in the previous section regarding aggregate scores apply here as well (albeit to a regressor and not to the response). Table 7 shows the report of the model fit from the paper, where *Subject* is 1 for an observation with an overt subject, *ORScoreZ* is the z-transformed orality score, *YearZ* is the z-transformed year the respective document was dated at, and *Macroregion* is Spain or Non-Spain. The random effect *Text* has 37 levels for 37 documents. Henceforth, we shall call this the *Paper model*.

We confirm the correctness of our reconstruction of the model reported in the paper by looking at the quote from Walkden et al. (2023, 558) (Footnote 14 omitted) and verifying that the provided numbers match those from our reconstruction of the model. Highlighting was added by us. Obviously, we have managed to set up our model exactly as the authors did for their paper.[20]

[20]After double and triple checking, we conclude that $p < 0.03$ for ORScoreZ:YearZ as reported in the

Table 7: The Paper model is a multilevel binomial model with a logit link. N=3773, AIC=2548.2, conditional R^2=0.2063, marginal R^2=0.1311. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels (s=0.31).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(0, 0, Spain)	-2.55	9.24	<0.0001	***
ORScoreZ		1.03	3.82	0.0001	***
YearZ		0.05	0.33	0.7435	
Macroregion	Non-Spain	0.63	1.96	0.0495	*
ORScoreZ:YearZ		-0.44	2.12	0.0342	*

> However, this trend also holds for Spain, which contradicts our sociolinguistic predictions. Since the orality effect discussed in section 4.3 may be at play in the bar charts, it is necessary to run a mixed-effects model to confirm any relationships across countries and centuries. A mixed model using glmer from the lme4 package (Bates et al. 2015) in R (R Core Team 2021) was run to find any such underlying patterns. The model had pronoun realisation (**3,773 total tokens**) as the binary-dependent variable; macroregion (Spain vs. Non-Spain), orality and year (both z-scored) as fixed effects; and document ID as a random effect. Country was originally included as a fixed effect but had too many levels and was replaced with macroregion. The interaction between year and orality was significant (**β=-0.44**, SE=0,21, **p<0.03**), where the negative coefficient means that for every year that passes, the effect of orality on overtness is lessened. Thus, there is an underlying diachronic effect, but as we can see from Figure 3, it is a small effect. Macro-region was also found to be just significant (**β=0.63**, SE=0,32, **p<0.05**), favouring overtness in every country but Spain. This result explains the apparent rise we saw for Spain in Figure 3 as just a product of orality: the later Spanish texts score higher in ORSCORE.

A very overt problem that *any* reviewer should have spotted is that in this section of the paper, p-values are given as *less than* (<), wheras in the report of Study 1, they were given as *equal to* (=), see our Section 2.1.1. Clearly, the values for both studies are rounded to two decimal points, but why this discrepancy? Also, the phrase *the orality effect discussed in section 4.3 may be at play in the bar charts* is extremely odd as the orality effect was at play in the language community (*if* it was), not in a plot in a paper from 2023. Also, *it is necessary to run a mixed-effects model to confirm any*

paper is a minor lapse.

relationships across countries and centuries is incorrect as such a necessity does not exist. There is never a theoretical necessity to use a random effect. see Schäfer (2020) for a brief introduction and Gelman and Hill (2006, 245–47) for a summary of the inconsistencies in such assumptions about when to use random effects.

Most importantly, brute-force NHST inference is used once again: less than 0.05 leads to a positive inference, greater than 0.05 leads to a negative or no inference. In Study 2, the model is even less informative than in Study 1, however, as we will show presently.

2.2.2 Alternative models

Again, we begin our reanalysis by comparing the Paper model to a *Null model* (not reported in tabular form), which is a model that only has a fixed intercept and a random intercept for the text. In other words, we allow the intercept to vary between groups of data points defined by the texts they occur in, and no other explanatory variables are included. The Null model is a multilevel binomial model with a logit link. N=3773, AIC=2562.62, conditional R^2=0.1835, marginal R^2=0. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels (s=0.74). The AIC of the Paper model is lower. The marginal R^2 of the Null model is of course 0, as there are no first-level fixed effects, but even the conditional R^2 of the Paper model is minimally higher than that of the Null model. All in all, it does not appear even from this simple evaluation as if the orality score, the macroregion, and the date of the document explain much of the variance in the data, as clearly indicated by the R^2 measures and the prediction accuracy.

A second alternative model we would like to entertain is the *Country-Genre model*, see Table 8. It drops the alleged main explanatory regressor (orality), the secondary one (year), and it specifies the country and the genre as fixed effects instead. Contrary to the claim in the paper, the standard maximum likelihood estimator converges without any additional tweaks, and the Country regressor does not have too many levels at all.[21]

We see in Table 9 that the Country-Genre model is also slightly worse than the Paper model. It is unclear, however, why the authors claim that the country regressor was impossible to add for technical reasons. The estimator has no trouble estimating the coefficients, they just don't reach the nominal NHST significance level of 0.05.

We proceed to estimating the parameters of a more complex model. The *Full-Country model* reported in Table 10 is the Paper model with orality and year as regressors but with country instead of macroregion and additionally genre as fixed effects.

[21] Even if it did indeed have too many levels for reliable fixed-effect estimation, a random effect (nesting the document ID) could and should have been used, see Gelman and Hill (2006); Schäfer (2020).

Table 8: The Country-Genre model is a multilevel binomial model with a logit link. N=3773, AIC=2566.93, conditional R^2=0.1853, marginal R^2=0.0465. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels (s=0.56).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(Spain, Nonliterary)	-2.79	7.61	<0.0001	***
Country	Bolivia	0.32	0.64	0.5212	
	Colombia	-0.12	0.28	0.7797	
	Dominican Republic	0.02	0.04	0.9669	
	Panamá	0.28	0.59	0.5531	
Genre	Literary	0.27	0.86	0.3885	
	Supplemental	1.31	2.53	0.0115	*

Table 9: Model comparison for the Country-Genre model

Name	AIC	R2_conditional	R2_marginal
SubjectCountryGenreModel	2566.931	0.1852888	0.0465148
SubjectPaperModel	2548.202	0.2063281	0.1310736

Table 10: The Full-Country model is a multilevel binomial model with a logit link. N=3773, AIC=2555.6, conditional R^2=0.2097, marginal R^2=0.1406. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels (s=0.29).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(0, 0, Spain, DOC)	-2.62	8.54	<0.0001	***
ORScoreZ		1.12	3.75	0.0002	***
YearZ		-0.06	0.33	0.7401	
Country	Bolivia	0.96	2.05	0.0400	*
	Colombia	0.68	1.61	0.1067	
	Dominican Republic	0.39	0.99	0.3213	
	Panamá	0.51	1.30	0.1924	
Genre	Literary	0.20	0.78	0.4367	
	Supplemental	0.28	0.56	0.5785	
ORScoreZ:YearZ		-0.58	2.45	0.0143	*

Table 11: Model comparison for the Full-Country model

Name	AIC	R2_conditional	R2_marginal
SubjectFullCountryModel	2555.599	0.2097230	0.1405899
SubjectPaperModel	2548.202	0.2063281	0.1310736

Table 12: The Full-No-Genre model is a multilevel binomial model with a logit link. N=3773, AIC=2552.27, conditional R^2=0.2107, marginal R^2=0.1391. The prediction accuracy on the training data (at a probability threshold of 0.5) is 0.89 given a base rate of 0.89. The only random effect is Text with 37 levels (s=0.3).

Term	Level	Coefficient	Sigma	P	Sig.
Intercept	(0, 0, Spain)	-2.52	9.26	<0.0001	***
ORScoreZ		1.15	3.89	0.0001	***
YearZ		-0.04	0.23	0.8175	
Country	Bolivia	0.99	2.14	0.0321	*
	Colombia	0.76	2.02	0.0437	*
	Dominican Republic	0.40	1.02	0.3081	
	Panamá	0.53	1.36	0.1741	
ORScoreZ:YearZ		-0.56	2.38	0.0173	*

The Full-Country model performs slightly better than the Paper model in terms of the R^2, as shown in Table 11. It should, however, have become clear by now that the slightly varying conditional R^2 coefficients are merely artefacts due to the shifting marginal R^2 coefficients. Additionally, we observe that country has only one level (Bolivia) that is sufficiently distinct from the intercept value (Spain) to reach the nominal significance level. Concluding from the data presented in the paper that "overtness [is favoured] in every country but Spain" is thus not warranted, even under an NHST umbrella. If we drop genre, the picture changes again with the *Full-No-Genre model*, see Tables 12 and 13.

We see that model specification is not an exact art. Dropping genre as a regressor leads to Columbia also reaching the nominal significance level. Based on NHST logic,

Table 13: Model comparison for the Full-No-Genre model

Name	AIC	R2_conditional	R2_marginal
SubjectFullNoGenreModel	2552.271	0.2106704	0.1390568
SubjectPaperModel	2548.202	0.2063281	0.1310736

it is now Spain, the Dominican Republic, and Panamá versus Bolivia and Columbia. Regarding the R^2, the various model specifications do not seem to change by any considerable margin. In other words, it does not seem to matter which regressors we include, the results are always middling at an almost exactly identical level. In the next section, we show why, focussing on the Paper model as one example of the many meaningless models one can specify for the data.

2.2.3 Interpretation of the models

As explained in the discussion of Study 1, we check the probability estimates made by the Paper model for the data used to estimate its parameters. First, we examine the correct binary predictions using a 0.5 probability threshold. The model predicts a null subject in 3773 cases and a pronominal subject in 0 cases. In other words, it predicts that there should never be an overt subject. This is due to the obviously uninformative regressors and the high base rate. The base rate (no overt subject) is at 88.55%. To see whether the Paper model could make better binary predictions with a lower probability threshold for binary decisions, we plot the predictions of the Paper model and the Null model next to each other as violin plots in Figure 4.

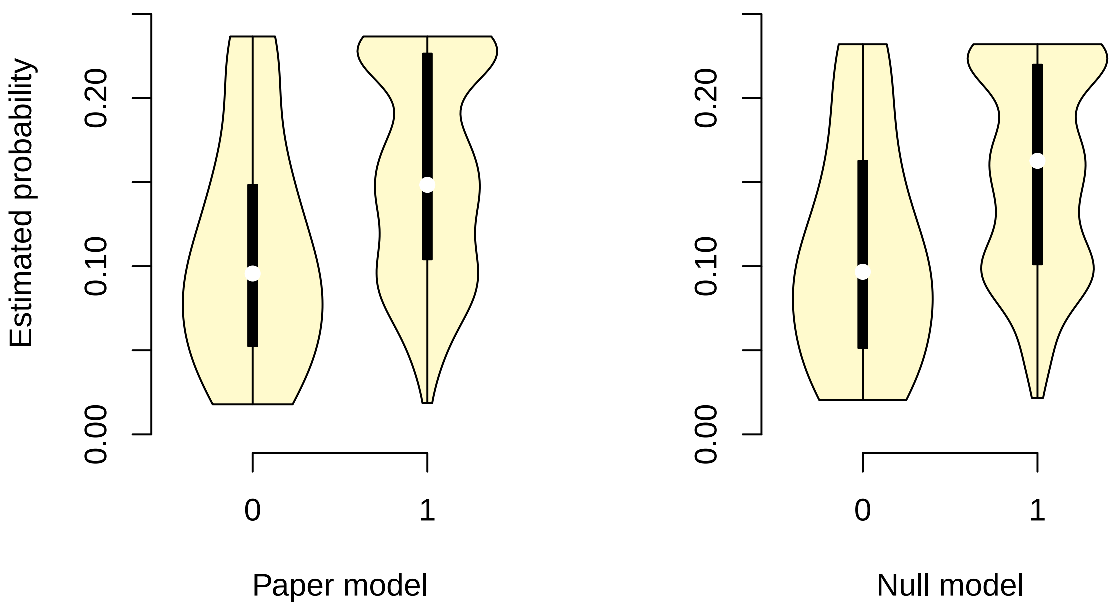

Figure 4: Probabilities estimated by the Paper model and the Null model; only for data used to estimate the model parameters and grouped by their actual outcome

Even for the cases with overt subjects in the observed data, the model estimates a maximum probability of under 0.25. Most strikingly, at least judging from the distribution of the predictions the Paper model estimates almost exactly the same probabilities as the null model. In other words, orality, time, and country (or macroregion) are by and large irrelevant. Reaching the significance level is almost exclusively an effect of the relatively simple model structure, minute differences correlated with the regressors, and the relatively large sample size. Alls points of criticism against NHST apply. Figure 5 shows the percentage of correct binary predictions of the model at different thresholds. The percentage rises quickly until the point where 100% minus the base rate percentage has been reached, which is indicative of a model that does nothing but predict the modal category, reaching the base rate in the best case.

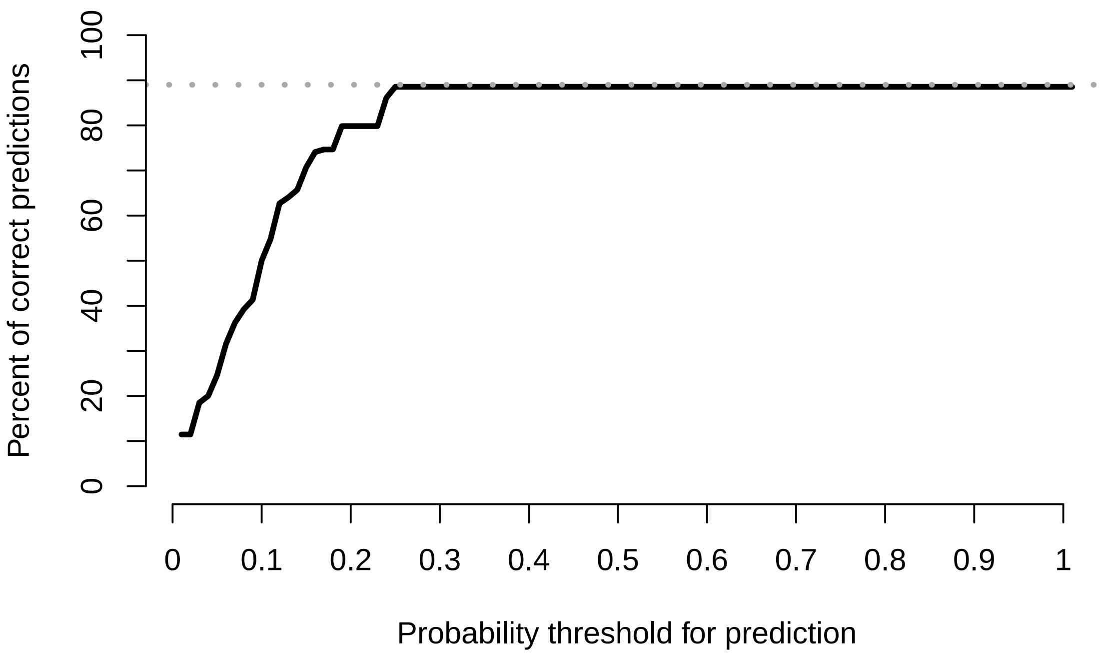

Figure 5: Percent of binary predictions based on the Paper model at varying thereshholds; only for data used to estimate the model parameters and grouped by their actual outcome

In Figure 6, we plot the Precision and Recall curves for the Paper model. Since the model does not predict any probabilities above 0.24, both precision and recall cannot be higher than 0 past this threshold.

Finally, and most revealing perhaps, we plot the probabilities predicted by the Paper model against those predicted by the Null model in Figure 7 in the style of a correlation plot. The plot shows which probability the two models estimate for each of the 3,773 observed data points, each data point being plotted as a dot. If the two models predicted exactly the same probabilities, the dots should be aligned along a straight line. Indeed, we see dots mostly aligned along a straight line with negligible deviations. The models contain virtually the same information, and the mean absolute difference between the predictions by the two models is 0.0069, the maximal difference is 0.045.

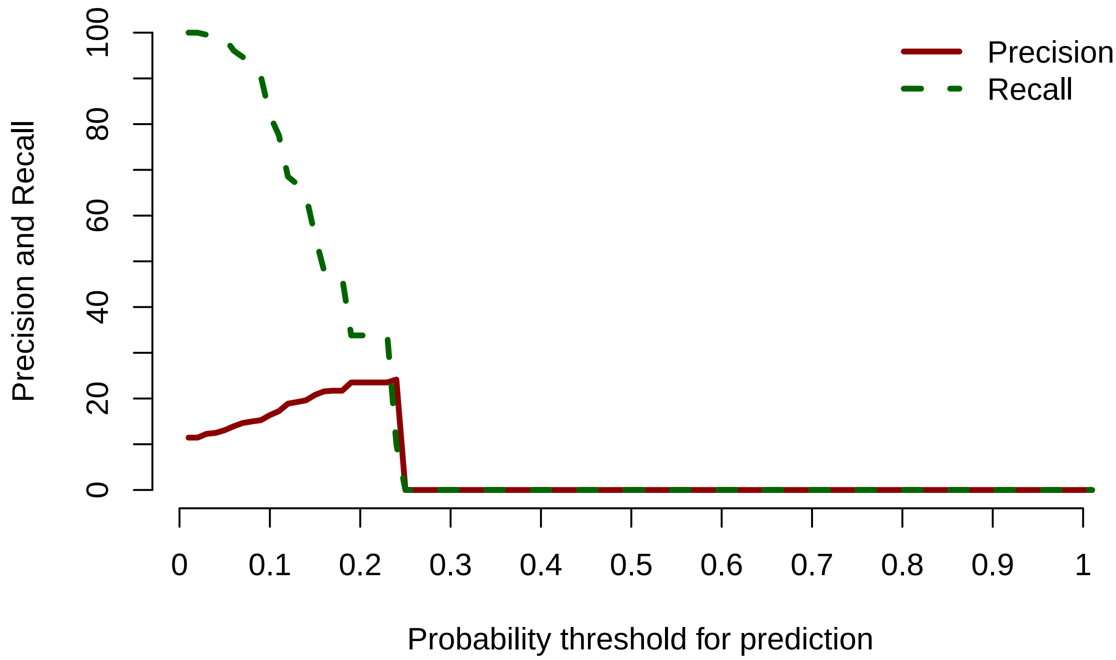

Figure 6: Precision and Recall for the Paper model at varying theresholds; only for data used to estimate the model parameters and grouped by their actual outcome

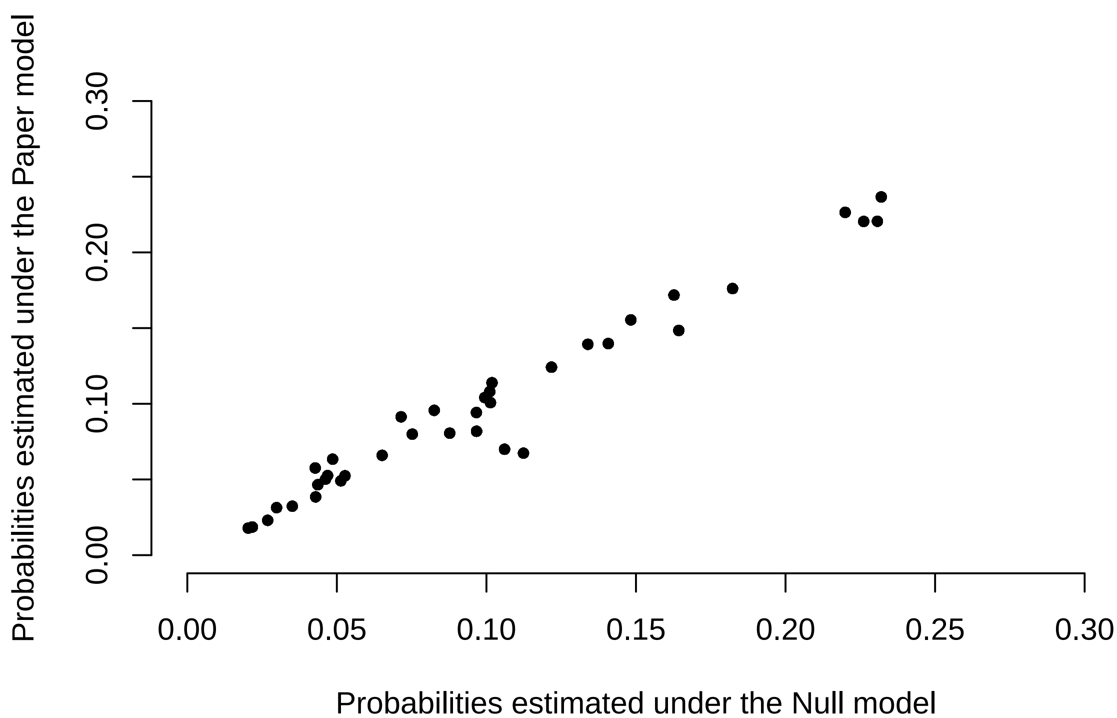

Figure 7: Comparing the predictions of the Paper model and the Null model

Furthermore and most crucially, we see only 37 dots, not 3,773. This is because the model makes exactly the same predictions for each data point from the individual texts, and the dots are therefore printed on top of each other. The model demonstrates (weakly) that individual medieval English texts differ from each other, nothing more. It does not at all "confirm any relationships across countries and centuries" (see above). These problems are not at all hard to pinpoint with full access to the data, and it is astonishing that the analysis went on to be submitted in paper form. We whole-heartedly support publishing negative findings, and hence the data themselves deserve publishing. However, negative findings should be clearly published as such, and not as corroborations of a massive and unclear network of hypotheses, especially given problems of underdetermination discussed in Section 1.

Study 3 reported in Walkden et al. (2023) is highly preliminary, and we therefore do not discuss it. The additive models used in the analysis have their own set of problems, and a discussion would take up too much space in the present paper. In the last section, we will discuss what all this means for substantive hypotheses put forward in Walkden et al. (2023) and the field in general.

3 Putting data to good use

3.1 Failure mode

We hope to have convinced readers that the data presented in Walkden et al. (2023) does provide strong evidence in favour of the claims put forward by the authors. However, the authors clearly think they do. How did this happen? As we showed in Section 1, the authors are looking for support of the Trudgill's theory of language change. This theory, we argued, consists of a huge and mostly implicit conjunction of substantive and auxiliary hypotheses. From these hypotheses, the authors derived (verbally, not using any strict logic) particular hypotheses about language changes in England and the Hispanopohone world. These particular hypotheses along with many additional implicit auxiliary hypotheses were added to the conjunct of hypotheses already assumed. From the particular hypotheses the authors implicitly derived statistical hypotheses (about the occurrence probabilities of overtly inflected pre-nominal elements in certain regions and times in England as well as the occurrence probability of subject drop in Spanish-speaking countries and regions). These statistical hypotheses were converted to Null hypotheses of the form "*Variable V does not have an influence on the respective probability.*", which were tested as parts of complex statistical models. From a rejection of these hypotheses using discredited NHST logic, positive inferences were made all the way back through the network of hypotheses to the effect of "*Trudgill's theory is correct.*" and "*The method proposed by us is effective in corroborating Trudgill's theory.*"

It should be obvious that this is a risky approach. Not only are two major positive inferences attempted, but the whole problem of underdetermination is completely ignored. Also, the amount of evidence (measured in the number of linguistic examples used in the statistical is minute compared to the complexity of the network of hypotheses. Unfortunately, the mechanical application of NHST and some incorrect analyses of the data (in th form of unweighted aggregations) invalidate the rejection of the weak Null hypotheses, and hence there is no inference to be made at all, neither a modest one nor an exceptional one. We can only speculate why the errors were not detected by the authors, the main author, the editors, and the reviewers. Instead of ending on such an uncomfortable note, we'd like to suggest three remedies to avoid such results to be published in the future. None of these is original in the sense that we did not come up with them. Other areas have similar problems, and many researchers have had very good ideas to remedy these problems.

3.2 Remedy 1: disciplined inference and better theories

We need to be modest about our inferences and acknowledge the network of hypotheses and the intrinsic underdeterminism in linguistics. Testing theories is a piecemeal exercise, and we should be aware that any corpus study or experiment that we report should focus on the most specific subset of hypotheses from the network. It would be a major advantage if our theories made more precise predictions. Ideally, theories should at least predict a numerical range for the expected measurements in a study. This would certainly sharpen our particular and statistical hypotheses to be tested. Alternatively, numerical predictions can also be obtained from experimental traditions, which only works if the exact same thing is measured over and over again. This allows us to conduct meta-analyses and establish numerical expectations (see Vasishth 2015 for an example from psycholinguistics).

If possible at all, numerical expectations should be used to test statistical hypotheses using Neyman-Pearson logic (Neyman and Pearson 1933; Neyman 1937, 1957) or extensions thereof such as severity (Mayo and Spanos 2006). These allow us to make valid positive inferences beyond mere rejection of uninformative Null hypotheses, but they usually require predictions about the numerical range we expect the results to land in.

Finally, researchers need better training in data analysis and statistical inference. We have seen many situations where postgraduates in linguistics were simply sent off to a one-week summer school and expected to return fully competent in the application of statistical methods. Many are also self-taught and rely on case-by-case internet searches to solve their problems. Often, dissertation supervisors appear to be not an ounce more competent than their candidates in the application of statistical methods. To the extent that this is not just an inadmissible impression from our episodic memory, it is indeed problematic.

3.3 Remedy 2: open data as part of the review process

Editors and reviewers must receive as much relevant information as possible in order to spot problems with statistical models. Also, a look at the data should be a required part of the review process, and reviewers should be competent to spot errors with ease. While there are clues in Walkden et al. (2023) as published, the real problems are only revealed through a reanalysis of the data. In our own experience, reviewers (even for corpus linguistics journals and journals with an empirical focus) rarely spot errors in the design of studies or the statistical analysis of the data. This might be because they have no access to the data, do not examine the data, or lack the skills to examine the data. Rather, they tend do demand more in-depth theoretical discussion, which more often than not does not make the paper's contribution to scientific progress any more sound or relevant.

A related issue is the lack of best-practice standards for linguistics journals. As it stands, most journals will accept any submission that reports a GLM or any other type of statistics, and it is the authors' choice which type of quality assurance they apply. As authors, we find this nerve-wrecking, especially as some reviewers are not shy to demand the quality assurance that suits their personal taste, which makes preparing publications rather difficult. A journal that has no best-practice guidelines for statistical methods should reject any papers applying those methods (unless the paper specifically makes an original methodological contribution, for example by introducing a new method altogether).

3.4 Remedy 3: negative findings an pre-registration

Universities and funding agencies require researchers to publish as many results as possible, and postgraduate students are under immense pressure to make themselves known by publishing in a frenzy. At the same time, it is very difficult to publish negative results. It would be demanding to re-write Walkden et al. (2023) along the following lines and still make it sound interesting to reviewers: "*We did not find evidence for Trudgill's theory in our data, but we'll keep looking.*"

Pre-registration is an attempt to alleviate this problem. Researchers submit a paper describing the motivation for a study and the design of the study *without having executed the study* and thus without having spent the time and money. The paper is accepted or rejected based on the submission, and whatever the results, the paper is published if the study was executed as designed. This makes negative results publishable and should allow researchers to be more critical with their own data. However, while we would welcome journals offering pre-registration, we think it does not solve too many problems. As negative results would not receive nearly as many citations as positive results, not much might be gained.

In closing, we'd like to point out that philology used to be a field where careful close readings of texts were performed in order to gain insights about their linguistic and cultural relevance. It was not bound by the scientific method and had no interest in data analysis and statistics. A major reason used to be that there are just not enough textual data covering all relevant times and places in order to do reliable statistical analysis. In many cases, we wonder whether this has really changed considerably, and why many researchers appear fixated on quantitative analyses where traditional philological analyses might be in order. If, however, quantitative analyses are not performed with the required rigour, and if theories are not taken seriously as monstrous underdetermined networks of intertwined hypotheses, nothing will be gained in a quantitative turn.

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